Quiz1

Keep Editing This Quiz

This is a preview of the published version of the quiz

Started: Sep 16 at 11:51am

Quiz Instructions

The quiz will be conducted **in person, in class**. Remote access is not permitted.

The quiz is **closed-note**.

The use of **online resources**, including search engines and generative AI, is **strictly prohibited.**

Only the Canvas quiz page should be open during the quiz. Opening any other tabs will be considered an act of academic dishonesty, and the student will be escorted out and reported.

Do not share the quiz with students who are absent, as they may need to take a make-up quiz.

Questions
Question 1
Question 2
Question 3
Question 4
Question 5
Question 6
Question 7
Question 8

Time Running: Hide Time
0 Minutes, 0 Seconds

Question 1 — 0.5 pts

low

Which of the following kernels is high pass, low pass, or neither?

(i) $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$ (ii) $\begin{bmatrix} \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \end{bmatrix}$ (iii) $\begin{bmatrix} -\frac{1}{9} & -\frac{1}{9} & -\frac{1}{9} \\ -\frac{1}{9} & \frac{8}{9} & -\frac{1}{9} \\ -\frac{1}{9} & -\frac{1}{9} & -\frac{1}{9} \end{bmatrix}$ → high

high neith lo

(i) [Choose]

(ii) [Choose]

(iii) [Choose]

Question 2 — 0.5 pts

What is the value of the pixel highlighted in the output image, filtered by a **3x3 box filter**?

image $f[\cdot,\cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

output $h[\cdot,\cdot]$

	0	10	20	30	30	30	20	10	
	0	20	40	60	60	60	40	20	
	0	$\boxed{}$							

$\frac{270}{9}$

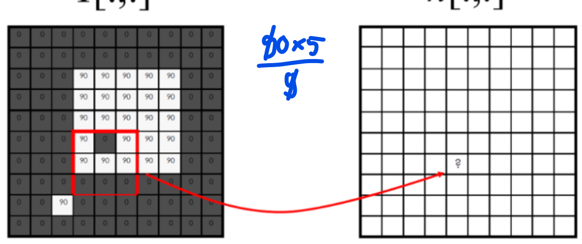
$= 30$

30

Question 3 — 0.5 pts

What is the value of the pixel marked as "?" after applying the 3x3 **median filter**?

$I[.,.]$ $h[.,.]$



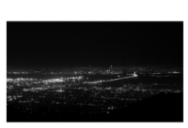
$\frac{90 \times 5}{9}$

50 → 90

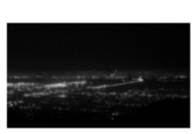
Question 4 — 0.5 pts

Given the input image below, identify if a high-pass or a low-pass has been used to achieve the output images.

Input image:



Output image 1:



low

Output image 2:

high

Image 1 [Choose]

Image 2 [Choose]

Question 5 — 1 pts

How can aliasing be avoided when sampling a signal? Select **all** the apply.

- [x] By over-sampling the signal at the Nyquist frequency
- [ ] By smoothing the signal to remove high frequency components
- [ ] By over-sampling at half the maximum frequency of the signal
- [ ] By over-sampling image twice the maximum frequency of the signal

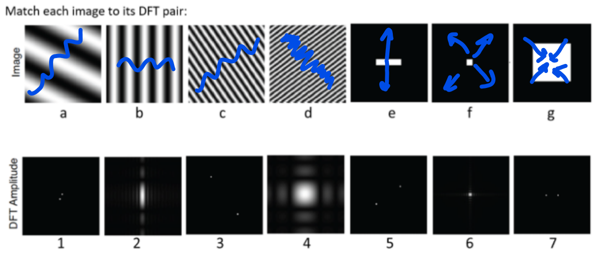
Question 6 — 1 pts

Given the transformations below, which one(s) can be used to reconstruct the original image? Select **all** the apply.

- [ ] Laplacian pyramid
- [ ] Gaussian pyramid
- [ ] ~~Blurred image using~~ a Gaussian kernel
- [ ] Fourier transform

Question 7 — 2 pts

Match each image to its DFT pair:



a [Choose] 1

b [Choose] 7

c [Choose] 5

d [Choose] 3

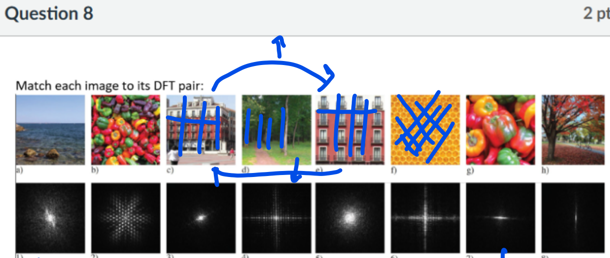
e [Choose] 2

f [Choose] 4

g [Choose] 6

Question 8 — 2 pts

Match each image to its DFT pair:



h f g e b c d a

a [Choose] 8

b [Choose] 5

c [Choose] 6

d [Choose] 7

e [Choose] 4

f [Choose] 2

g [Choose] 3

h [Choose] 1

Question 9 — 1 pts

Suppose image $h = f * g$ is the output of **convolving** image f with filter g, and that f and h have **DFT transforms** shown below. What is the value of the pixel highlighted in DFT(h)?

DFT(f) DFT(g) DFT(h)

0	0.5	0.5	0
0.5	1	1	0.5
0.5	1	1	0.5
0	0.5	0.5	0

+

0	0	0	0
0	1	1	0
0	1	1	0
0	0	0	0

+

	?		

0.5 ?

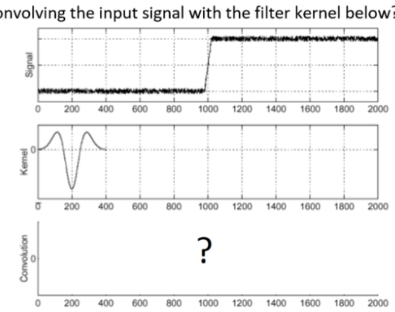
What is the output of the convolving the filter kernel below to the input signal?

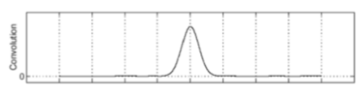
Question 10

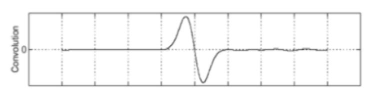
0.5 pts

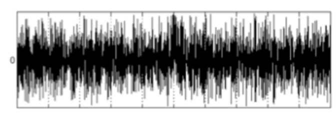
What is the output of convolving the input signal with the filter kernel below?

Lap



a) 

b) 

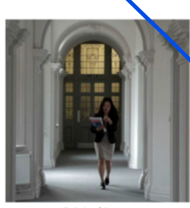
c) 

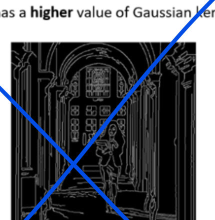
- b
- c
- a

Question 11

0.5 pts

Which Canny edge detector has a **higher** value of Gaussian kernel σ?


Original image


A


B

- A
- B

Quiz saved at 11:53am

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